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AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 305 — Past Questions & Materials

by Renance

Course: CSC 305

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

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Paper 1 of 1 — CSC 305

Level 300 • First Semester • PastQuestion

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AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
2020/2021 SESSION FIRST SEMESTER DEGREE EXAMINATION
CSC 305: ARTIFICIAL INTELLIGENCE

Instruction: Attempt Question 1 Any Other 3 Questions Time Allowed: 2hrs

1. a) Briefly describe Alan Turing's test used as an assessment to test the intelligence of machines. (4mks)
b) Discuss how the components of an Expert System allow Human Experts to have an edge over them. (5mks)
c) Trace briefly the history of Artificial Intelligence. (7mks)
d) Describe any four forms of Learning in Artificial Intelligence. (4mks)
e) How "artificial" is *Intelligence* in Artificial Intelligence? (5mks)
f) Translate this sentence into first order predicate logic:
"Every male who admires the Metaverse community dislikes every other male who the Metaverse community loves." (10mks)
2. a) State 5 common Artificial Intelligence techniques practically applied in Drone technology. (5mks)
b) Describe how Deep Learning, Machine Learning and Artificial Intelligence are related. (5mks)
c) What characteristics of humans can make artificial agents intelligent? (5mks)
3. a) Name five modern practical examples of weak AI. (5mks)
b) Must Intelligent Agents be foolproof? Discuss. (5mks)
c) State the components to develop a smart CAPTCHA computer. (5mks)
4. a) Discuss the challenges facing Robotics technology in recent times. (4mks)
b) State the factors that a rational agent should depend on at any given time. (4mks)
c) Discuss the future of Artificial Intelligence with regards to Augmented Reality (AR) and Virtual Reality (VR). (7mks)
5. a) What is Artificial Intelligence? Discuss how it can be applied on Google Search on a daily basis. (6mks)
b) What is "Planning" in knowledge, reasoning and planning Artificial Intelligence Systems? (5mks)
c) Define an Agent, and explain why it is useful. (4mks)
6. a) Translate this sentence into first order predicate logic: **"If Tesla's Model Pi phone is a Starlink-compatible smartphone and all Starlink-compatible smartphones have solar charging, then Model Pi is self-charging"**. (10mks)
b) State the properties of environment in intelligent agents. (5mks)

AMBROSE ALLI UNIVERSITY EKPOMA
DEPARTMENT OF COMPUTER SCIENCE
2016/2017 SESSION FIRST SEMESTER DEGREE EXAMINATION
CSC 305: ARTIFICIAL INTELLIGENCE

Instruction: Attempt Question 1 Any Other 3 Questions Time Allowed: 2hrs

1. a) State the systems that categorize Artificial Intelligence agents into areas that humans do better. (4mks)
 b) Describe the methodology that makes an artificial agent act humanly. (6mks)
 c) Write a basic environment simulator *Procedure* program that would give each agent its percept, gets an action from each agent, and then updates the environment. (7mks)
 d) Briefly explain the components of an Expert System. (3mks)
 e) Explain why a Depth-First Search AI technique is never a complete search procedure; whereas a Breadth-First Search AI technique is complete. (5mks)
 f) Translate this sentence into first order predicate logic:
 "Every hostel flat is such that either they have only females, or they have exactly two males, or they have no student." (10mks)
2. a) With the aid of a diagram, how would you illustrate a complete utility-based agent? (7mks)
 b) Describe the roles of knowledge representations in solving problems. (5mks)
 c) State the logic behind a) Hill climbing, b) Best-First Search, c) Breadth-First Search. (3mks)
3. a) Is *Turing Test* the only legitimate test of machine intelligence? Discuss. (5mks)
 b) Differentiate Informed & Uninformed search. Give examples. (5mks)
 c) Briefly describe the history of Artificial Intelligence. (5mks)
4. a) What are the advantages and disadvantages of an expert system? (6mks)
 b) When does an agent act intelligently? (4mks)
 c) State how you would convert a sentence to Conjunctive Normal Form (CNF) during Knowledge Reasoning and Planning of an agent. (5mks)
5. a) List the AI approaches that have aided in human reasoning and of the nature of intelligence. (5mks)
 b) What is "Planning" in knowledge, reasoning and planning Artificial Intelligence Systems? (5mks)
 c) What are the roles of LISP and PROLOG in Artificial Intelligence? (5mks)
6. a) Define the set of sentential formulas of first-order logic. (5mks)
 b) List a few of the task domains of an Artificial Intelligence. (5mks)
 c) State the factors that a rational agent should depend on at any given time. Explain any One with an example. (5mks)

- a) Assuming you were given a language $\{a,b,c\}$. Using RE, show that all strings contain at least one occurrence of each symbol in the given language (6 marks)
- (b) Identify the properties of primitive regular expression and state the algorithm used to composed additional regular expression from its primitive (6 marks)
- (c) With relevant example show the closure properties of automata (4 marks)
- (d) Using computational theory in automata systems, prove that if $yx^2=0$, then $x=0$ (4 marks)

CNS/FCS/MED/15/18378

Propeller

AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
2019/2020 SESSION FIRST SEMESTER DEGREE EXAMINATION
CSC 305: ARTIFICIAL INTELLIGENCE

Instruction: Attempt Question 1 Any Other 3 Questions Time Allowed: 2hrs

1. a) Briefly describe any method used as an assessment to test the intelligence of machines. (4mks)
 b) Discuss how the components of an Expert System allow Human Experts to have an edge over them. (5mks)
 c) The crop yield in Esan-West Local Government is depreciating because farmers are not able to detect certain diseases in crops during the early stages. Can Artificial Intelligence be used for disease detection in crops? If yes explain. (7mks)
 d) State any 4 applications of CAPTCHA. (4mks)
 e) How Artificial is Intelligence in Artificial Intelligence? (5mks)
 f) Translate this sentence into first order predicate logic:
"Combat Drones only hit precise targets that are not allies." (10mks)
2. a) State 5 common Artificial Intelligence techniques practically applied in Drone technology. (5mks)
 b) Describe how Deep Learning, Machine Learning and Artificial Intelligence are related. (5mks)
 c) What characteristics of humans can make artificial agents intelligent? (5mks)
3. a) Speech recognition software and driverless cars are becoming more useful and feasible in recent times. State and explain the types of AI that can categorise them. (5mks)
 b) Must Intelligent Agents be foolproof? Discuss. (5mks)
 c) State the components to develop a smart CAPTCHA computer. (5mks)
4. a) List the disadvantages of Robot applications. (4mks)
 b) State the factors that a rational agent should depend on at any given time. (4mks)
 c) Discuss the future of Artificial Intelligence with regards to Drones and Vertical Agriculture. (7mks)
5. a) What is Artificial Intelligence? Discuss how it can be applied on Google Search on a daily basis. (6mks)
 b) What is "Planning" in knowledge, reasoning and planning Artificial Intelligence Systems? (5mks)
 c) Define an Agent, naming 3 examples. (4mks)
6. a) Translate this sentence into first order predicate logic: **"If my pitbull dog's eyes are red and all red-eye dogs are dangerous, then my dog is dangerous"**. (10mks)
 b) List any 5 language characteristics of First Order Predicate Logic. (5mks)

AMBROSE ALLI UNIVERSITY, EKPOMA
FACULTY OF NATURAL SCIENCES
DEPARTMENT OF COMPUTER SCIENCE
2015/2016 FIRST SEMESTER DEGREE EXAMINATIONS

CSC 305: ARTIFICIAL INTELLIGENCE

Instructions: Answer Question ONE and Any other THREE Questions Time: 2Hrs

- What is AI? (2 marks)
- Differentiate an agent function and an agent program (3 marks)
- Explain with the aid of a schematic diagram the model based reflex agent (3 marks)
- Humans developed AI systems by introducing into them every possible intelligence they could, for which the human themselves now seem threatened. Discuss. (3 marks)
- Explain your understanding of Iterative Deepening Depth-first search (3 marks)
- Indicate the performance of algorithmic complexities based on the criteria below, by completing the table: (6 marks)

Criterion	Breadth First	Depth First	Didirectional	Uniform cost	Iterative Deepening
Time	$O(b^d)$	$O(b^m)$	$O(b^{d/2})$	$O(b^{d+1})$	$O(b^d)$
Space	$O(b^d)$	$O(b \cdot m)$	$O(b^{d/2})$	$O(b^d)$	$O(b \cdot d)$
Optimality	Yes	No	Yes	Yes	Yes
Completeness	Yes	No	Yes	Yes	Yes

- What is A.I. technique? (5 marks)
 - Differentiate between inductive reasoning and deductive reasoning (5 marks)
 - Differentiate between human and machine intelligence (5 marks)
 - Enumerate any five broadly common and prospering research areas in the domain of A.I. (5 marks)
- State the requirements of the Brute-force search strategies (4 marks)
 - Give a comprehensive description of the following and also state their advantages and disadvantages:
 - Breadth-First search (i) Depth-First search (ii) Uniform-cost search (6 marks)
 - Differentiate between A* search and Greedy Best First Search (5 marks)
 - Explain your understanding of the Eliminated Amending and also state the algorithm for performing it. (5 marks)
- What is a task environment? How is it specified? (5 marks)
 - List the properties of task environment. (4 marks)

- (f) Write a function for the update operation. (3 marks)
- (g) Write the learning reports. (2 marks)
- (h) Your friend is a developer, explain the final model of learning agents. (3 marks)

- 29(a) State five applications of fuzzy logic. (5 marks)
- (a) Discuss the three different levels of ambiguity of the FL. (3 marks)
- (b) What do the following terms depend on? (4 marks)
- (i) Fuzzy logic (ii) Fuzzy logic (iii) Fuzzy logic (iv) Fuzzy logic
- (c) Explain your understanding of the following terms: (8 marks)
- (i) Fuzzy logic (ii) Fuzzy logic (iii) Fuzzy logic (iv) Fuzzy logic

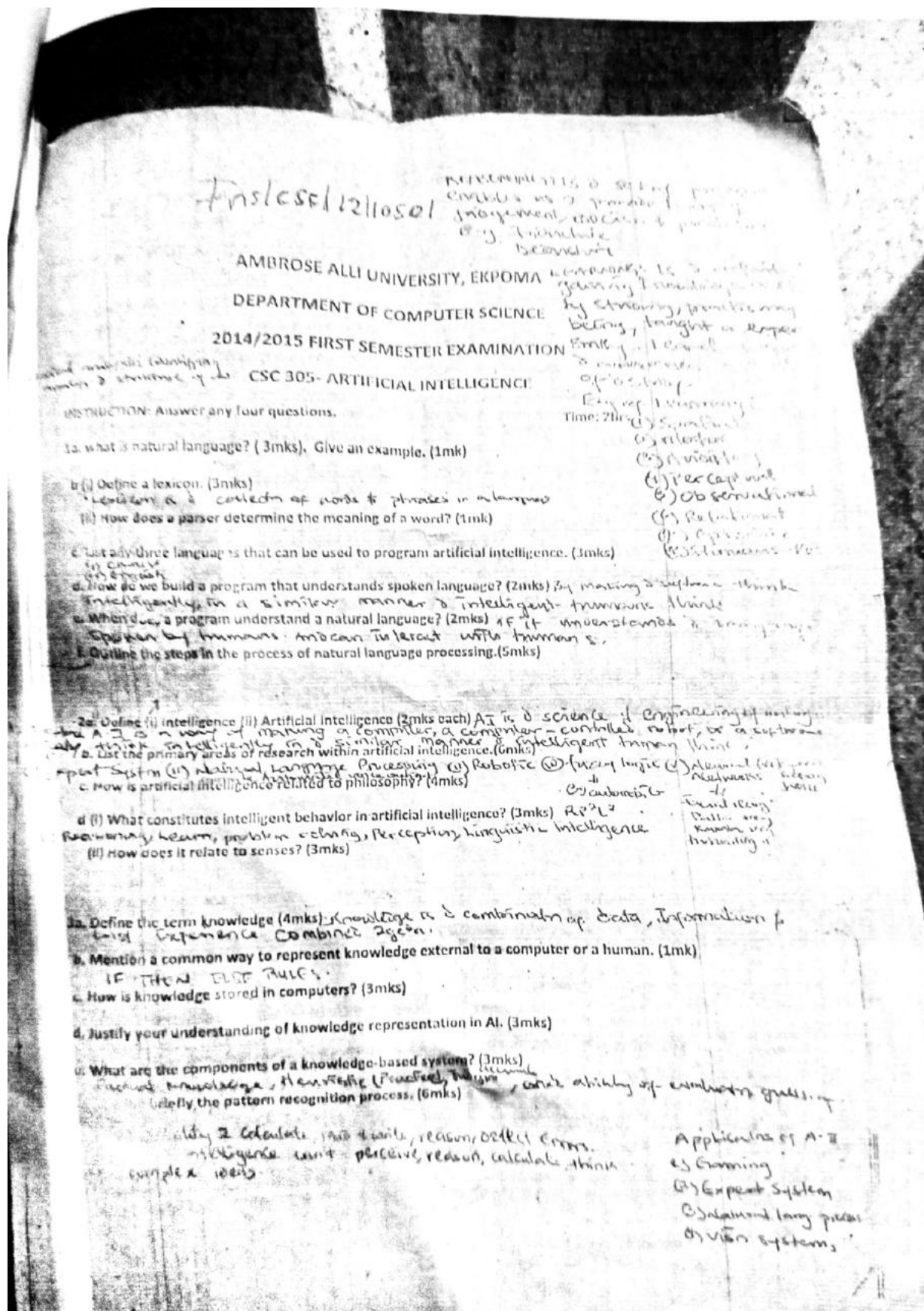
Consider the Markov Decision process below. Actions have non-deterministic results. Taking an action in a state results in different states with some probability. There are two actions out of each, D for development and R for research. Consider the following table which ultimately care only about money expected for any transition resulting in a state.

State	S1	S2	S3	S4
Research	0	100	25	5

Suppose you start with state S1 and perform the following actions:

- Action R, New State: S3
- Action D, New State: S2
- Action R, New State: S1
- Action R, New State: S3
- Action R, New State: S4
- Action D, New State: S2

- (a) Assume $V(s)$ for all $s = S1, S2, S3$ and $S4$ is initialized to 0. Update $V(s)$ for each of the states using the Temporal Difference Algorithm. (12 marks)
- (b) What would your policy be at the end of training in (a) above? (3 marks)



Computer Vision: Automatically extracts, analyzes & comprehends useful information from an array of images.
 It has 2 components: an environmental component of computer vision: camera, illumination, power supply, display software, power supply, display software, power supply, display software.
 Functions of Computer Vision: Detect, object, track, face detection, estimate position.

(i) Define (a) robot (b) vision (3mks each)
 (ii) Define (i) a robot (ii) vision (3mks each)
 (iii) Define (i) a robot (ii) vision (3mks each)

c. What are the two common denominator of perception that can be incorporated into robots. (5mk)

d. List the various categories of sensory perception of computer vision. (4mk)

e. Outline the impacts of robots. (4mk)

Aspects of Relap: Good Impact
 Psychological Aspect: Good Impact
 Socioeconomic Aspect: Good Impact
 Cognitive Aspect: Good Impact
 Computational Intelligence (CI) an agent (links each)

b. What are the goals of an intelligent agent? (3marks) *CS 2 Create Expert System*
 c. Why is an intelligent agent appropriate for its circumstances and its goals? (3marks) *CS 2 Implement Knowledge Representation*
 d. Write informed notes on any two of the following:
 i. Turing Test (4marks each)

[illegible]

a. Define the term expert system and explain its components. (5 marks)
 b. How is expert system different from conventional computer systems? (5 marks)
 c. Give a typical example of an expert and explain its functions in human analogy. (5 marks)

1. From your answer in question 6c above, name the principle of the operation of the device. The device is dependent on the relation of frequency to the physical properties of the material.

2. List the limitations of the device. The limitations of the device are: (a) the device is only useful for measuring the frequency of a signal, (b) the device is only useful for measuring the frequency of a signal, (c) the device is only useful for measuring the frequency of a signal. What is the principle of the device? The principle of the device is that the frequency of a signal is proportional to the physical properties of the material.

Point can be defined as a Mechanical/Physical agent
by the Computer to Carry Out Specific Task.
Physical agents acting to read words and turn it

- (3) Mechanical constraints.
- (4) Power electrical components.
- (5) Environmental requirements.

Can be defined as the inf...

the existing compartmentalized models of human behavior by viewing
 the state of being a rational agent of having judgments of judgment
 (or) personal decisions

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